Alkane

21. $CH_3-CH_2-CH_2-CH_3 \xrightarrow[HBr]{AlCl_3} \text{Product}$.

Product in above reaction is

(a) $CH_3-\underset{\underset{Br}{|}}{CH}-CH_2-CH_3$

(b) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(c) $\underset{\underset{Br}{|}}{CH_2}-CH_2-\underset{\underset{CH_3}{|}}{CH_2}$

(d) All of these

22. Which of the following statements is not true for ethane

(a)It can be chlorinated with chlorine

(b)It can be catalytically hydrogenated

(c)When oxidised produces CO_2 and H_2O

(d)It is a homologue of iso-butane

23. Petroleum refining is

(a)Distillation of petroleum to get different fractions

(b)Obtaining aromatic compounds from aliphatic compounds present in petroleum

(c)Cracking of petroleum to get gaseous hydrocarbons

(d)Purification of petroleum

24. The chemical added to leaded petrol to prevent the deposition of lead in the combustion chamber is

(a) Iso-octane

(b) Ethylene dibromide

(c) Tetraethyl lead

(d) Mercaptan

(e) *n*-Heptane

25. In the commercial gasolines, the type of hydrocarbons which are more desirable is

(a) Branched hydrocarbon

(b) Straight-chain hydrocarbon

(c) Linear unsaturated hydrocarbon

(d) Toluene

26. Which of the following is not formed by the reaction of Cl_2 on CH_4 in sunlight

(a) $CHCl_3$

(b) CH_3Cl

(c) CH_3CH_3

(d) $CH_3CH_2CH_3$

27. Which of the following has the highest boiling point

(a) Neopentane (b) *n*-butane

(c) *n*-heptane (d) Isobutane

28. Which gives CH_4 when treated with water

(a) Silicon carbide

(b) Calcium carbide

(c) Aluminium carbide

(d) Iron carbide

29. Which of the following does not react with PCl_5

(a) CH_3OH (b) CH_3COOH

(c) CH_3CHO (d) C_2H_6

30. Which of the following compounds is insoluble even in hot concentrated H_2SO_4

(a) Ethylene (b) Benzene

(c) Hexane (d) Aniline

31. A reaction between methyl magnesium bromide and ethyl alcohol gives

(a) Methane (b) Ethane

(c) Propane (d) Butane

32. Methane and ethane both can be obtained in single step from

(a) CH_3I (b) C_2H_5I

(c) CH_3OH (d) C_2H_5OH

33. Paraffin wax is

(a) Ester

(b) Alcohol

(c) Unsaturated hydrocarbon

(d) Saturated hydrocarbon

34. The number of possible enantiomeric pairs that can be produced during monochlorination of 2-methylbutane is

(a) 2 (b) 3

(c) 4 (d) 1

35. Petroleum consists mainly of

(a) Aliphatic hydrocarbons

(b) Aromatic hydrocarbons

(c) Aliphatic alcohols

(d) None of these

36. Petroleum ether can be used as

(a) Solvent for fat, oil, varnish and rubber

(b) As a fuel

(c) Both (a) and (b)

(d) None of these

37. Which of the following are produced from coaltar

(a) Synthetic dyes

(b) Drugs

(c) Perfumes

(d) All the three

38. In alkanes, the bond angle is

(a) 109.5^o (b) 109°

(c) 120° (d) 180°

39. In the preparation of alkanes; a concentrated aqueous solution of sodium or potassium salts of saturated carboxylic acid are subjected to

(a) Hydrolysis

(b) Oxidation

(c) Hydrogenation

(d) Electrolysis

40. Halogenation of alkanes is an example of

(a) Electrophilic substitution

(b) Nucleophilic substitution

(c) Free-radical substitution

(d) Oxidation

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